Green's XVA: mathematics and framework

9 October 2026 · David Kerr Kelly

Andrew Green, *XVA: Credit, Funding and Capital Valuation Adjustments* (Wiley, 2015), derives every adjustment from one argument: hedge the trade and both defaults with a self-financing portfolio, write the PDE that forces, and apply Feynman–Kac. Each XVA is then a spread times a discounted expected exposure, integrated over the trade's life, and the funding strategy you assume decides which terms appear. Equations below were transcribed from the scanned book; the FVA, MVA and KVA results were checked against the page images, and apparent misprints are flagged.

Notation

Every symbol used anywhere in this document, grouped by role. Conventions:

- Time t runs from today to maturity T. The letters s and u are integration variables over time.
- All expectations are under the risk-neutral measure Q, with the money-market account as numeraire.
- $V^+ = \max(V, 0)$ is the positive part of V and $V^- = \min(V, 0)$ is the negative part.
- An XVA with a negative value is a cost to the bank.

Market and trade

Symbol	Meaning
S	Price of the underlying (a stock in Green's derivation; an oil forward in the worked example)
μ, σ	Real-world drift and volatility of S
W	Brownian motion driving S
γ_S	Dividend yield on S
q_S	Repo rate at which a stock position is financed
r	Risk-free short rate; in practice the collateral (OIS/RFR) rate
H(S)	Trade payoff at maturity T
V	Risk-free value of the trade: the value with no default risk, funding cost or capital cost
$\hat{V}$	All-in value of the trade, including every adjustment
U	Total valuation adjustment, $U = \hat{V} - V$
EPE, ENE	Expected positive exposure $E[V^+]$ and expected negative exposure $E[V^-]$ at a future date

Credit and default

Symbol	Meaning
τ_B, τ_C	Default times of the bank (B) and the counterparty (C)
J_B, J_C	Default indicators: 0 before default, jump to 1 at default
λ_B, λ_C	Risk-neutral hazard rates (instantaneous default intensities) of bank and counterparty
R_B, R_C	Recovery rates of bank and counterparty
$\Phi(\tau > t)$	Survival probability to time t, $\exp(-\int_0^t \lambda\, du)$
P_C, r_C, q_C	Zero-recovery counterparty bond; its yield; its repo rate. These give $\lambda_C = r_C - q_C$
P_1, P_2	Two bonds issued by the bank, of different seniority
R_1, R_2	Recovery rates on P_1 and P_2
r_1, r_2	Yields on P_1 and P_2, with $r_i - r = (1 - R_i)\lambda_B$ (no bond–CDS basis)
M_B, M_C	Close-out amount agreed on bank or counterparty default (standard: $M = V$)
g_B, g_C	Value the bank actually ends up with on its own or the counterparty's default

Funding, collateral, margin and capital

Symbol	Meaning
X, r_X, s_X	Variation margin (collateral) held; the rate it earns; $s_X = r_X - r$
I, $\tilde{I}$	Initial margin posted and received (segregated, cannot be reused)
r_I, s_I, $r_{\tilde{I}}$	Rate earned on posted IM, its spread $s_I = r_I - r$, and rate paid on received IM
r_F, s_F, R_F	Bank's unsecured funding rate, spread $s_F = r_F - r = (1 - R_F)\lambda_B$, and funding-bond recovery
K	Regulatory capital held against the trade
γ_K	Cost of capital (shareholders' required return)
φ	Fraction of capital that can be used to fund the trade ($0 \le \varphi \le 1$)
γ_E, E	Effective tax rate; taxable cash flow (TVA)
ψ, ξ	Fraction of counterparty default hedged; market price of default risk (warehousing)
K^U, K^R	Capital with no CVA hedge; capital relief from a full hedge

Hedge portfolio and operators

Symbol	Meaning
δ	Units of S held in the hedge
α_C, α_1, α_2	Units of P_C, P_1 and P_2 held (negative = issued)
Π	Value of the hedge portfolio
ΔV̂_B, ΔV̂_C	Jump in V̂ at bank or counterparty default: g_B − V̂ and g_C − V̂
ε_h	Hedge error: what the hedge fails to cover at the bank's own default
𝒜_t	Pricing operator, 𝒜_t f = ½σ²S² ∂²f/∂S² + (q_S − γ_S) S ∂f/∂S
D(t, s)	Survival-adjusted discount factor exp(−∫ₜˢ (r + λ_B + λ_C) du)
𝔼_t	Risk-neutral expectation conditional on information at time t

The framework: semi-replication in five steps

Green adopts Burgard and Kjaer's semi-replication and extends it, one cost at a time, from CVA to FVA, MVA, KVA and TVA. Every chapter repeats the same five steps; only the hedge portfolio, the funding constraint and the close-out change.

Step 1 — Split the value. Write the all-in value as the risk-free value plus an adjustment:

$$\hat{V}(t,S) = V(t,S) + U(t,S)$$

Here V is the textbook price, with no default, funding or capital, and U collects every XVA.

Step 2 — Write the dynamics. The underlying diffuses. The counterparty bond and the bank's two bonds earn their yields and jump down on default:

$$dS = \mu S\,dt + \sigma S\,dW, \qquad dP_C = r_C P_C\,dt - P_C\,dJ_C, \qquad dP_i = r_i P_i\,dt - (1-R_i)\,P_i\,dJ_B \quad (i=1,2)$$

In words: when J_C jumps from 0 to 1, the zero-recovery counterparty bond loses its whole value. When J_B jumps, the bank's bond i loses the fraction (1 − R_i). These two equations define the hazard rates:

$$\lambda_C = r_C - q_C, \qquad (1-R_i)\,\lambda_B = r_i - r$$

Step 3 — Set the close-out and build the hedge. On default the trade is closed out at M (standard choice M_B = M_C = V), net of collateral X. The bank then holds:

$$g_C = R_C\,(V-X)^+ + (V-X)^- + X, \qquad g_B = (V-X)^+ + R_B\,(V-X)^- + X$$

On the counterparty's default the bank recovers only R_C of what it is owed. On its own default it pays only R_B of what it owes.

The hedge holds δ units of S, α_C counterparty bonds and α_1, α_2 own bonds. Each position is financed through a cash account (stock at repo q_S with dividends γ_S; counterparty bond at repo q_C; collateral earning r_X). The **funding constraint** says the trade, net of collateral, is financed entirely by issuing the bank's own bonds:

$$\hat{V} - X + I + \alpha_1 P_1 + \alpha_2 P_2 - \phi K = 0$$

The I and φK terms enter only in the MVA (Ch. 10) and KVA (Ch. 13) extensions; set them to zero for CVA and FVA.

Step 4 — Remove the risks. Require the trade plus hedge to be riskless, dV̂ + dΠ = 0. That fixes:

$$\delta = -\frac{\partial \hat{V}}{\partial S}, \qquad \alpha_C P_C = g_C - \hat{V}, \qquad \epsilon_h = (g_B - \hat{V}) - \left[(1-R_1)\alpha_1 P_1 + (1-R_2)\alpha_2 P_2\right]$$

- δ removes the market risk dS.
- α_C removes the counterparty jump dJ_C.
- The bank cannot freely trade its own default, so the own-default jump is hedged only as far as the funding constraint allows. Whatever is left is the hedge error ε_h. That residual is where funding cost comes from; it is derived, not assumed.

Step 5 — Read off the XVA. The risk-free value solves the textbook PDE:

$$\frac{\partial V}{\partial t} + \mathcal{A}_t V - rV = 0, \qquad V(T,S) = H(S), \qquad \mathcal{A}_t f = \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} + (q_S - \gamma_S)\, S \frac{\partial f}{\partial S}$$

Subtracting it from the PDE for V̂ leaves a linear PDE for the adjustment. Each right-hand-side term is one source of cost or benefit:

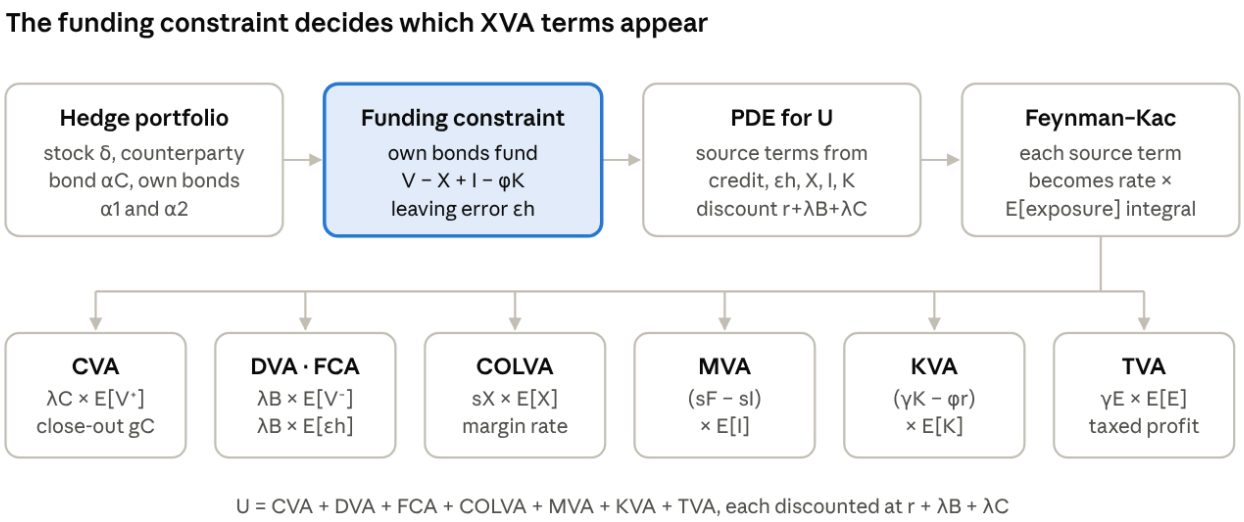
$$\frac{\partial U}{\partial t} + \mathcal{A}_t U - (r + \lambda_B + \lambda_C)\, U = -\lambda_C\,(g_C - V) - \lambda_B\,(g_B - V) + \lambda_B\,\epsilon_h + s_X X - s_I I + r_{\tilde{I}}\,\tilde{I} + (\gamma_K - \phi\, r)\, K, \qquad U(T,S) = 0$$

Feynman–Kac turns a PDE of the form ∂U/∂t + 𝒜_tU − kU = f, with U(T) = 0, into the expectation U(t) = −∫ₜᵀ 𝔼_t[e^{−∫ₜˢ k du} f(s)] ds. Here k = r + λ_B + λ_C, so every adjustment is an integral of a rate times a survival-adjusted, discounted expectation:

$$U(t) = -\int_t^T D(t,s)\, \mathbb{E}_t\big[f(s)\big]\, ds, \qquad D(t,s) = e^{-\int_t^s (r+\lambda_B+\lambda_C)\, du}$$

Source term f	Becomes	Rate × exposure (standard close-out)
−λ_C (g_C − V)	CVA	(1 − R_C)λ_C × E[(V − X)⁺]
−λ_B (g_B − V)	DVA	(1 − R_B)λ_B × E[(V − X)⁻]
λ_B ε_h	FCA (funding cost)	depends on the funding strategy, Ch. 9
s_X X	COLVA	collateral spread × E[X]
−s_I I (with funding of I)	MVA	(s_F − s_I) × E[I]
(γ_K − φr) K	KVA	cost of capital × E[K]

The general result is U = CVA + DVA + FCA + COLVA + MVA + KVA, plus TVA when tax is added (Ch. 14). Because k contains λ_B + λ_C, survival of both parties is built into every term, not just CVA.



Green's derivation chain · 4 steps, 6 XVA terms

Change the funding strategy and the hedge error ε_h changes with it. That is why the same trade carries DVA + FCA under one strategy and a symmetric FVA under another.

CVA and DVA: expectation and replication (Ch. 3)

Green derives CVA twice, by expectation and by replication, and shows that they agree. The replication version is the one that generalises.

By expectation. The value is the no-default cash flows plus the recovery value at default, where V_τ is the close-out value:

$$\mathrm{CVA}(t) = \mathbb{E}\big[(1-R)\,(V_\tau)^+ \mid \mathcal{F}_t\big] \qquad (3.11)$$

$$\mathrm{CVA}(t) = (1-R)\sum_{i=0}^{n-1}\big[\Phi(\tau > t_i) - \Phi(\tau > t_{i+1})\big]\,\mathbb{E}\Big[e^{-\int_0^{t_i} r\,du}\,(V_{t_i})^+\Big] \qquad (3.13)$$

The bilateral version uses first-to-default indicators. With independent defaults, each period's default probability is weighted by the other party's survival:

$$\mathrm{BCVA} = \mathbb{E}\big[(1-R_C)(V_{\min})^+ 1_{\tau_C<\tau_B<T}\big] + \mathbb{E}\big[(1-R_B)(V_{\min})^- 1_{\tau_B<\tau_C<T}\big] \qquad (3.44)$$

By replication. Hedge the stock with δ and the counterparty jump with a zero-recovery bond, α_C P_C = g_C − V̂. Use repo-financed hedges and risk-free close-out; in Chapter 3 the bank funds at r. The adjustment PDE then gives:

$$\partial_t U + \mathcal{A}_t U - (r+\lambda_C+\lambda_B)U = -\lambda_C(g_C - V) - \lambda_B(g_B - V) + s_X X \qquad (3.65)$$

$$\mathrm{CVA} = -(1-R_C)\int_t^T \lambda_C(s)\,e^{-\int_t^s (r+\lambda_C+\lambda_B)\,du}\,\mathbb{E}\big[(V(s)-X(s))^+\big]\,ds \qquad (3.70)$$

$$\mathrm{DVA} = -(1-R_B)\int_t^T \lambda_B(s)\,e^{-\int_t^s (r+\lambda_C+\lambda_B)\,du}\,\mathbb{E}\big[(V(s)-X(s))^-\big]\,ds \qquad (3.71)$$

$$\mathrm{COLVA} = -\int_t^T e^{-\int_t^s (r+\lambda_C+\lambda_B)\,du}\,\mathbb{E}\big[s_X(s)\,X(s)\big]\,ds \qquad (3.72)$$

Allocation. EPE = E[max(Σ α_i V_i, 0)] is homogeneous of degree one in the trade weights, so Euler allocation gives trade contributions that sum exactly to the netting-set total (3.84–3.89). Green computes them in the same Monte Carlo pass by bumping each weight (3.93). For incremental CVA he recommends one simulation over both the old and new sets with common random numbers.

Worked example (5-year GBP swap, CDS 300bp, bank 100bp, recoveries 40%): unilateral CVA 87,907; bilateral CVA 84,746; DVA −49,054; BCVA 35,691 (Tables 3.1 and 3.3; the printed DVA total is garbled and was re-summed from the rows).

Green's positions.

- DVA restores price symmetry but cannot be hedged, since a bank cannot trade its own CDS. Basel excludes it, and many dealers prefer to treat it as a funding benefit.
- He does not recommend a Gaussian copula between the two defaults for bilateral CVA.

Credit inputs, analytic CVA, collateral and wrong-way risk (Ch. 4–7)

Default probabilities. Default is reduced-form, with survival P(τ > T) = exp(−∫λ). The CDS legs include accrued premium on default. The breakeven spread equates premium and protection legs, and the protection leg is:

$$V_{\text{prot}}(t_0) = (1-R)\sum_{k=1}^{p}\frac{\lambda_k}{\lambda_k + r_k}\Big[B(t_0,u_{k-1})P(\tau > u_{k-1}) - B(t_0,u_k)P(\tau > u_k)\Big] \qquad (4.14)$$

Green bootstraps a piecewise-constant hazard rate on IMM-aligned segments and gives a minimum admissible spread (4.23) to test whether a curve can build, rather than adjusting quotes by hand. He warns that CVA carry jumps at the quarterly CDS roll, because the hazard segments are redefined.

Proxy spreads. He prefers the cross-sectional model, log S = Σ A_ij log m_j, with global, sector, region, rating and seniority factors fitted per tenor (4.25–4.26). He notes CDS spreads embed CVA-capital relief, estimated at 20–50% of the price (Kenyon–Green 2013b). He proposes a capital-relief leg (4.24) to back out capital-adjusted hazard rates.

Analytic swap CVA. Exposure at each date is a European swaption on the remaining swap, so CVA is a strip of swaptions weighted by default probabilities:

$$\text{CVA}^{\text{swap}}(t_0) = (1-R)\sum_{i=0}^{n-1}\left[\Phi(\tau > t_i) - \Phi(\tau > t_{i+1})\right] V_{\text{PS}}(t_0,t_i,K) \qquad (5.8)$$

This is useful as a Monte Carlo benchmark, but only if the swaptions are priced in the same model, smile included. Caplets and FX forwards reduce to one option times the default probability (5.16, 5.21).

Collateral and margin period of risk. Collateral enters through the close-out, (V − X)⁺. The simplest threshold model is a call spread, E[(1−R)·min(max(V,0), H)], but it ignores the margin period and understates exposure. With a cure period:

$$\text{CVA} = \mathbb{E}\big[(1-R_C)\left(V(\tau) - X(\tau - \Delta t_{\text{cure}})\right)^{+}\big] \qquad (6.37)$$

Green's practical recommendation is the lookback method: X(t) = max(V(t − Δt) − H, 0), which doubles the time steps (6.39). Direct daily collateral simulation is the most accurate but too costly over 30–40 years. Downgrade triggers are handled with rating-conditional EPE profiles (6.44–6.46).

Wrong-way risk. CVA becomes an expectation of exposure conditional on default at that time:

$$\mathrm{CVA}^{\mathrm{WW}} = (1-R)\int_t^T e^{-\int r}\,\mathbb{E}\big[(V(u))^+ \mid \tau = u\big]\,d\Phi(t,u) \qquad (7.4)$$

His distributional models are all changes of measure within one framework:

- **Gaussian copula** between the exposure distribution and default time (7.9).
- **Hull–White hazard-on-exposure,** λ = exp(a + b·P(t)), with Green's variant using max(P, 0) (7.10–7.13).
- **Glasserman–Yang,** where any path-weighting P_ij that respects the default marginals is admissible. The worst case is a linear programme, CVA_max = max Σ P_ij C_ij (7.17–7.22).

The alternative family is stochastic intensity with jumps (7.23). On a 30-year swap with 2% CDS, CVA moves from about £1.0m to £3.1m across the Hull–White b range, against £2.2m with no WWR (Table 7.1, Fig. 7.3). Green treats WWR correlation as a stress parameter, not a price.

CSA discounting and the collateral option (Ch. 8)

Green follows Piterbarg. With collateral X earning r_X and unsecured funding at r_F, the hedging argument gives a PDE in which the risk-free rate does not appear:

$$\partial_t V + \tfrac{1}{2}\sigma_S^2 S^2 \partial_S^2 V + (q-\gamma) S\,\partial_S V - r_F V = X\,(r_X - r_F) \qquad (8.29)$$

$$V(t) = \mathbb{E}_t\Big[e^{-\int_t^T r_X\,ds}\,V(T)\Big] - \mathbb{E}_t\Big[\int_t^T e^{-\int_t^s r_X\,du}\,s_F\,(V-X)\,ds\Big] \qquad (8.33)$$

With perfect collateral (X = V), the second term vanishes and the trade is discounted at the collateral rate. Since collateral earns overnight, that means the OIS curve (8.34). With no collateral, it is discounted at the funding rate, which is the "first-generation FVA" taken up in Chapter 9.

Cheapest-to-deliver option. If the poster can choose the collateral currency, the effective collateral rate is the best of the domestic rate and each foreign rate adjusted by the FX-swap basis:

$$P_{\mathrm{ctd}}(t,T) = \exp\Big\{-\int_t^T \big(c_d + \max(c_f + r_{d,f} - c_d,\,0)\big)\,ds\Big\} \qquad (8.36)$$

This assumes deterministic rates. With stochastic rates, Green uses Piterbarg's Jensen-type approximation (8.51). Green flags sticky collateral (no substitution rights) and non-cash collateral with haircuts as unresolved.

FVA: Burgard–Kjaer semi-replication (Ch. 9)

This chapter is the core of the book. FVA is not assumed; it appears as the cost of the part of the hedge that cannot be completed, namely the bank's own default.

Set-up. Hold the stock, a zero-recovery counterparty bond, and two own bonds P_1 and P_2 with different recoveries (different seniorities). Each own bond's yield satisfies r_i − r = (1 − R_i)λB, which assumes no bond–CDS basis. The derivative net of collateral is financed by own-bond issuance:

$$\hat{V} - X + \alpha_1 P_1 + \alpha_2 P_2 = 0 \qquad (9.30)$$

$$g_B = (V - X)^+ + R_B (V - X)^- + X, \qquad g_C = R_C (V - X)^+ + (V - X)^- + X \qquad (9.22,\ 9.23)$$

Hedging δ and α_C leaves a hedge error at own default, ε_h = ΔV̂_B − (1 − R_1)α_1P_1 − (1 − R_2)α_2P_2. The adjustment then solves:

$$\partial_t U + \mathcal{A}_t U - (r + \lambda_B + \lambda_C) U = s_X X - \lambda_C (g_C - V) - \lambda_B (g_B - V) + \lambda_B\, \epsilon_h \qquad (9.41)$$

$$U = \mathrm{CVA} + \mathrm{DVA} + \mathrm{FCA} + \mathrm{COLVA}, \qquad \mathrm{FCA} = -\int_t^T \lambda_B(s)\, e^{-\int_t^s (r + \lambda_B + \lambda_C)\, du}\, \mathbb{E}[\epsilon_h(s)]\, ds \qquad (9.43,\ 9.46)$$

The funding strategy decides the answer. How the bank splits its funding across the two bonds fixes ε_h, and therefore which terms appear:

Strategy	Own-bond holdings	Hedge error ε_h	Result
Perfect replication, two bonds	Long one seniority, short the other (9.50–9.51)	0	Bilateral CVA + DVA; no FVA. Green calls it unrealistic.
Semi-replication, no shortfall at own default	U funded with zero-recovery bonds; V with recovery-R_B bonds (9.52–9.53)	(1 − R_B)V⁺, a windfall to bondholders	CVA + DVA + FCA; DVA becomes the funding benefit
One bond, recovery R_F	All of V̂ funded at r_F = r + s_F (9.60–9.61)	Own default not hedged	Unilateral CVA + symmetric FVA; λB drops out

No shortfall at own default (checked against the page):

$$\mathrm{FCA} = -(1 - R_B) \int_t^T \lambda_B(s)\, e^{-\int_t^s (r + \lambda_B + \lambda_C)\, du}\, \mathbb{E}\big[V(s)^+\big]\, ds \qquad (9.57)$$

$$\mathrm{FVA} = \mathrm{DVA} + \mathrm{FCA} = -(1 - R_B) \int_t^T \lambda_B(s)\, e^{-\int_t^s (r + \lambda_B + \lambda_C)\, du}\, \mathbb{E}\big[V(s)\big]\, ds \qquad (9.58,\ 9.59)$$

With one bond, the PDE is discounted at the funding rate and g_B drops out (9.64–9.67):

$$\mathrm{CVA} = -(1 - R_C) \int_t^T \lambda_C\, e^{-\int_t^s (r_F + \lambda_C)\, du}\, \mathbb{E}[V^+]\, ds, \qquad \mathrm{FVA} = -\int_t^T s_F\, e^{-\int_t^s (r_F + \lambda_C)\, du}\, \mathbb{E}[V]\, ds$$

FVA splits into a cost on E[V⁺] and a benefit on E[V⁻] (9.68–9.70). This is the version most practitioners use, because the unhedgeable DVA disappears.

Close-out matters.

- With risky close-out, which includes XVA in the close-out amount, the equation becomes non-linear in U and needs regression plus iteration (9.77–9.82).
- With close-out at the survivor's funding value, CVA and FVA separate completely (9.83–9.96).

Double counting. Morini and Prampolini's loan example (9.2–9.9) shows that a borrower adding both a funding benefit and DVA counts its own credit twice. That is the formal reason FBA and DVA cannot both be booked in full.

Green's position on the FVA debate.

- Kenyon and Green prove that if participants face different holding costs on the same asset (Theorem 9.1), no single market-wide risk-neutral measure exists.

- Regulatory capital and funding change a dealer's cash flows. That breaks the Modigliani–Miller premise behind Hull and White's objection to FVA.
- Value is therefore entity-specific, and asymmetric between bank and corporate, or between two banks with different regulatory status (Tables 9.2, 9.3).
- A strong CSA leaves little residual FVA. A weak one (thresholds, infrequent calls) still needs it.

MVA and computing IM inside Monte Carlo (Ch. 10–11)

MVA by semi-replication (Green and Kenyon 2015).

- Add initial margin posted (I) and received (Ĩ) to the Chapter 9 portfolio. IM is segregated and cannot be rehypothecated.
- Posted IM must be funded by issuing bonds: V̂ − X + I + α_1P_1 + α_2P_2 = 0 (10.8).
- On default, received IM reduces the loss to the bank and posted IM increases the loss to the counterparty:

$$g_B = (V - X + I)^+ + R_B(V - X + I)^- + X - I, \qquad g_C = R_C(V - X - \tilde{I})^+ + (V - X - \tilde{I})^- + X + \tilde{I} \qquad (10.6,\ 10.7)$$

Using the no-shortfall strategy, ε_h = (1 − R_B)[V − X + I]⁺ (10.28), and Feynman–Kac gives U = CVA + FVA + COLVA + MVA (checked against the page):

$$\mathrm{CVA} = -(1 - R_C) \int_t^T \lambda_C(s)\, e^{-\int_t^s (r+\lambda_B+\lambda_C)\,du}\, \mathbb{E}\big[(V(s) - X(s) - \tilde{I}(s))^+\big]\, ds \qquad (10.31)$$

$$\mathrm{MVA} = -\int_t^T \big[(1 - R_B)\lambda_B(s) - s_I(s)\big]\, e^{-\int_t^s (r+\lambda_B+\lambda_C)\,du}\, \mathbb{E}\big[I(s)\big]\, ds \qquad (10.34)$$

MVA is the funding spread less the return earned on posted IM, applied to the expected IM profile. Received IM appears only in CVA and in COLVA, through its full yield r_Ĩ.

The computational problem. The expected IM profile E[I(s)] needs a VaR at every path and time step. Brute force is 10^4 swaps × 10^3 paths × 10^2 steps × 10^3 VaR scenarios = 10^{12} valuations, about 277 core-hours.

Longstaff–Schwartz Augmented Compression (LSAC). Green regresses portfolio value, not its moments, on the state variables:

$$\min_{\alpha} \sum_{\omega} \Big\| \sum_{m=1}^{n_f} \alpha_m f_m\big(O_i(\omega, t_k)\big) - F(\omega, t_k) \Big\|^2, \qquad V^j(\omega, t_k) \approx \bar{F}\big(\alpha_m, \hat{O}_i^{\,j}(\omega, t_k), t_k\big) \qquad (10.38,\ 10.40)$$

1. **Compress.** One regression function replaces the whole portfolio valuation, so cost is independent of the number of trades.
2. **Augment.** Each Monte Carlo state is shifted by a historical VaR shock before regressing, so the fit covers the tails the VaR needs. This also solves the t = 0 problem, where there is only one state.
3. **Revalue.** Apply all historical shocks to each state through the regression, sort the results and read off the percentile. Average across paths to get E[I].

Early-start Monte Carlo is the alternative augmentation when there is early exercise. A delta-gamma variant differentiates the regression function analytically.

Results (USD swap books, 99% 10-day VaR, 1,294 shocks from January 2007):

- VaR errors of a few basis points.
- About 100× faster than brute force at 10,000 swaps, on a single GPU.

Book (payer share)	Unsecured FVA (bp)	MVA (bp)
90% payer	115	53
50% payer (balanced)	0	2
10% payer	−113	56

MVA is roughly 40% of the unsecured FVA for directional books. Even a balanced book pays 2bp, against a pre-crisis bid-offer of about 0.25bp (Table 10.2).

Measure issue. Historical VaR shocks are P-measure, but E[I] is a Q-expectation. Green keeps the shocks fixed and exogenous, like a static stressed window.

The funding curve (Ch. 11). Green defines s_F = (1 − R_B)λB + γ_B, where γ_B is a liquidity term (11.4). He reviews the candidate curves:

- **Own CDS** is contaminated by CVA-capital hedging demand.
- **Bond spread** is what Burgard–Kjaer intend.
- **Bond–CDS basis** is too noisy to use as a liquidity measure.
- **Treasury transfer price** carries the treasury desk's own positions and bid-offer.

His preference is an optimal rolling-funding strategy, which is usually cheaper than term funding. For accounting, he argues for a market-consensus curve rather than the bank's own.

KVA: capital in the replication (Ch. 13)

Green and Kenyon put regulatory capital into the semi-replication as equity rented from shareholders. Holding capital K costs γ_K per unit of time:

$$d\bar{\beta}_K = -\gamma_K(t)\, K\, dt \qquad (13.1)$$

A fraction φ of capital may be used to fund the derivative, so the funding constraint becomes $\hat{V}$ − X + α_1P_1 + α_2P_2 − φK = 0 (13.2). Green's base case is φ = 0, which he thinks best reflects practice. Capital has no jump term: it neither changes the bank's recovery nor absorbs counterparty losses. The adjustment PDE gains two capital terms:

$$\partial_t U + \mathcal{A}_t U - (r + \lambda_B + \lambda_C)U = -(g_C - V)\lambda_C - (g_B - V)\lambda_B + \epsilon_h \lambda_B + s_X X + \gamma_K K - r\phi K \qquad (13.19)$$

Feynman–Kac gives U = CVA + DVA + FCA + COLVA + KVA (checked against the page):

$$\mathrm{KVA} = -\int_t^T \big(\gamma_K(s) - \phi\, r(s)\big)\, e^{-\int_t^s (r+\lambda_B+\lambda_C)\,du}\, \mathbb{E}\big[K(s)\big]\, ds \qquad (13.25)$$

KVA is the cost of capital, net of any funding benefit from capital, applied to the expected capital profile. It is discounted for both parties' survival, like every other term. An equivalent split moves the φr term into FCA (13.26–13.27).

Computing E[K(t)] (§13.4):

- **Standardised market risk** is essentially state-independent, so it simply rolls down the maturity table.
- **CEM, standardised and SA-CCR EADs** are computed as a "payout" in each Monte Carlo state.
- **IMM VaR** uses LSAC.
- **IMM CCR and advanced CVA capital** would need nested simulation; Green proposes regression instead.

Worked example. A 10-year GBP swap at 2.7%:

- Bank spread 100bp; counterparty spreads 30 to 750bp; recoveries 40%.
- Standardised capital throughout, CEM for EAD, 8% capital ratio, 10% cost of capital.

The table shows total XVA (CVA + DVA + FCA + KVA) in bp of notional, φ = 0 (Tables 13.3–13.5):

Trade	Unhedged	Back-to-back hedge under perfect CSA	Hedge sized for zero IR01
Pay fixed, AAA	−253	9	−13
Pay fixed, CCC	−341	−156	−249
Receive fixed, AAA	−325	−63	−87
Receive fixed, CCC	−587	−402	−512

- **Unhedged.** Market-risk KVA dominates. For pay-fixed AAA it is −262 out of the −274 KVA, against CCR −3 and CVA −9. It falls as credit worsens because of the survival discounting. Setting φ = 1 cuts KVA by about 25%.
- **Back-to-back hedge.** Market-risk capital goes to zero, but the hedge adds CCR and CVA capital, and the book's IR01 is not zero (0.61 to 2.39 for payers). The XVA terms themselves carry rate risk that the capital-optimal hedge leaves open.
- **Zero IR01.** Re-sizing the hedge to flatten IR01 brings back a small market-risk KVA (−17 to −67).

Green's conclusion is that KVA is the same order of magnitude as CVA, DVA and FVA. Chapter 13 assumes credit risk is perfectly hedged; Chapter 14 relaxes this.

Warehousing and TVA (Ch. 14)

This chapter (Kenyon and Green 2015) drops the assumption that counterparty default is fully hedged. The bank hedges a fraction ψ of the jump-to-default with the counterparty bond and prices the unhedged rest at its expected cost under the physical measure. Two things follow:

- Hedging earns capital relief: the capital charge becomes K^U − ψK^R (14.12).

- Profits are taxed, which adds a tax account (14.13).

$$\lambda_C^{\mathbb{P}} = \lambda_C(1-\xi), \qquad \tilde{\lambda}_C = \psi\,\lambda_C + (1-\psi)(1-\xi)\,\lambda_C \qquad (14.20,\ 14.21)$$

Here ξ is the market price of default risk. Every XVA is now discounted with the blended hazard λ̃_C, and two terms change:

$$\mathrm{KVA} = -\int_t^T e^{-\int_t^s (r+\lambda_B+\tilde{\lambda}_C)\,du}\,\mathbb{E}\Big[(\gamma_K - \phi r)\,(K^U - \psi K^R) + \lambda_B\,\epsilon_{B_K}\Big]\,ds \qquad (14.31)$$

$$\mathrm{TVA} = -\int_t^T e^{-\int_t^s (r+\lambda_B+\tilde{\lambda}_C)\,du}\,\mathbb{E}\Big[\gamma_E\,E(s) + \lambda_C(1-\xi)(1-\psi)\,\bar{\Delta}_E\Big]\,ds \qquad (14.32)$$

The taxable cash flow is E(t) = γ_K(K^U − K^R) + 𝟙_Tax·λB·ε_B (14.33). The indicator is 1 when the tax authority follows own-credit accounting.

Choosing ψ is an optimisation. Less hedging lowers CVA, because $\lambda^{\mathbb{P}} < \lambda$, but raises capital and taxable P&L. Part of every CDS spread reflects other banks' CVA-capital hedging, up to about 40% per Kenyon and Green (2013b). A bank that does not hedge therefore pays twice. The chapter gives no numerical results.

Portfolio KVA and capital allocation (Ch. 15)

Capital is a portfolio quantity, which creates two problems:

- Market-risk capital and the leverage ratio are set across all counterparties.
- The cost of capital depends on how much the bank holds, γ_K = γ_K(K) (15.50).

Green therefore writes the semi-replication for the whole bank. When counterparty J defaults, the bank receives the close-out on J and keeps the rest of the book: $\hat{V} = g_{CJ} + \hat{V}^{\{-J\}}$ (15.5). The PDE for U is then coupled across counterparties through $\hat{V}^{\{-j\}}$, ε_h and K.

Separability condition. U splits into counterparty pieces, U = Σ U_j, only if the hedge error and capital split additively. So capital must be allocated so that the pieces sum to the total:

$$K = \sum_j K_j, \qquad \mathrm{KVA}_j = -\int_t^T \big(\gamma_K(K,u) - r(u)\phi_j\big)\,e^{-\int_t^u (r+\lambda_B+\lambda_{C_j})\,ds}\,\mathbb{E}_t\big[K_j(u)\big]\,du \qquad (15.41,\ 15.56)$$

Allocation by capital type.

Capital	Allocation	Note
Market risk (standardised, VaR, SVaR, ES)	Euler	Degree-one homogeneous, so contributions sum to the total
Counterparty credit risk	Natural	Already computed per netting set
Standardised CVA capital, hedged	Pro rata on (M_j EAD_j − M_j^hedge B_j) (15.43)	The formula is not homogeneous, so Euler fails
Leverage ratio	Euler on the binding constraint (15.48–15.49)	Binds only when risk-based capital is below 3% of exposure

The split by counterparty is not unique and depends on the allocation scheme; only the total is invariant.

Algorithm 3: portfolio KVA by regression.

1. For each counterparty, simulate its own assets. Regress V_j, CVA capital, CCR capital and leverage exposure on the state variables (LSAC).
2. Run one global simulation. Compute market-risk capital there and sum the counterparty regressions to get the leverage-ratio capital.
3. Form total K and γ_K(K), and attribute market-risk and leverage capital to counterparties.
4. For each counterparty, run a second simulation with its K_j to get KVA_j. Portfolio KVA is Σ KVA_j.

Worked example: an uncollateralised oil hedge for a corporate producer (not in the book)

Neither Green nor Gregory works a commodity hedge, so this example builds one. It applies Green's integrals to the typical Macquarie trade: a corporate hedging oil with no CSA, with the bank offsetting the risk in margined futures.

Capital uses today's rules, SA-CCR and BA-CVA, not the CEM and old CVA charge in the book. All inputs are illustrative.

The result:

- With credit independent of oil, total XVA of USD 1.82m slightly exceeds the USD 1.69m margin in the price.
- Recognising right-way risk turns the trade profitable on the economics.
- Regulatory capital gives no credit for right-way risk, and it is the largest cost.

The trade and the inputs

Item	Value
Trade	Producer sells 100,000 bbl/month of Brent forward for 36 months at a fixed price; bank pays fixed K, receives monthly floating
Size	3.6m bbl, about USD 258m notional
Fixed price	K = USD 71.20/bbl: the par price less a USD 0.50/bbl bank margin
Curve and rates	Brent forwards 75·e^(−0.03T) (mild backwardation); flat USD rate r = 4%
Volatility	One-factor forward-curve model (Green 16.320) with Samuelson volatility σ(T − t) = 25% + 15%·e^(−(T−t)): about 40% at the front, 25% long-dated
Counterparty	BBB− producer: CDS 200bp, R_C = 40%, so λ_C = 3.33%
Bank	CDS 75bp, R_B = 40%, so s_F = 0.75%
Hedge	Bank sells Brent futures; initial margin about 8% of remaining notional value, funded at s_F
CCR capital	SA-CCR; oil supervisory factor 18%; α = 1.4; Basel 3.1 BBB corporate risk weight 75%; 8% ratio
CVA capital	BA-CVA reduced; energy sector, investment grade, risk weight 3%; DS = 0.65
Cost of capital	γ_K = 10%; φ = 0
Monte Carlo	60,000 antithetic paths, weekly steps

Step 1 — Value. With F(t, T_j) the forward for settlement month T_j, the bank's value is:

$$V(t) = Q \sum_{T_j > t} \left[F(t,T_j) - K \right] e^{-r\,(T_j - t)}, \qquad \frac{dF(t,T)}{F(t,T)} = \sigma(T-t)\,dW_t$$

At inception V(0) = USD 1,693,368, which is the bank's margin.

Because the curve is backwardated, the early settlements pay the bank and the later ones pay the producer. The expected value therefore turns negative after about six months: in effect the bank borrows from the producer. That is why FVA here is a small benefit, not a cost.

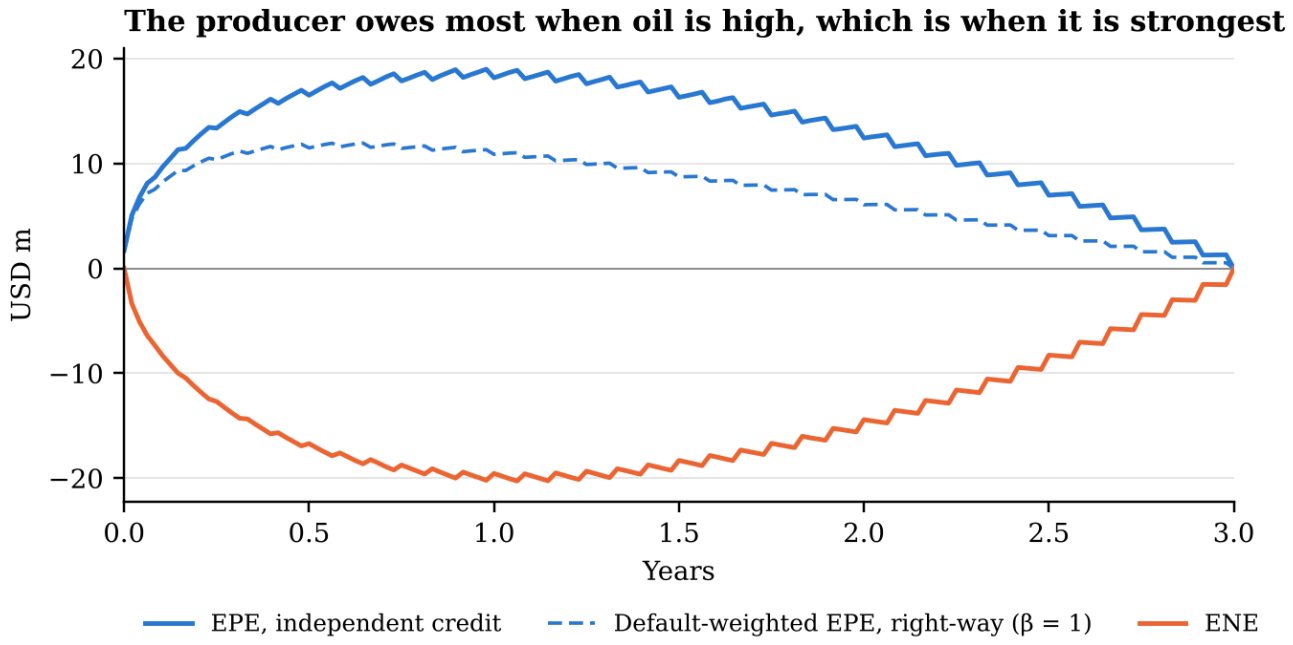
Step 2 — Exposure.

- The producer owes the bank when oil rises. Peak EPE is USD 19.0m (7% of notional) and average EPE USD 13.0m.
- The bank owes the producer when oil falls, with peak ENE of USD −20.3m.
- The exposure peaks around year one, where the remaining volume times the accumulated volatility is largest.

Step 3 — Right-way risk. A producer's credit improves when oil rises, which is exactly when it owes the bank. Following Green's path-weighting approach (Ch. 7), the counterparty's hazard rate on each path is scaled by the oil price. The weights are normalised so that the market-implied default probabilities are unchanged:

$$\lambda_C(t,\omega) = \lambda_C(t)\,\frac{\big(F(t)/F(0)\big)^{-\beta}}{\mathbb{E}\big[(F(t)/F(0))^{-\beta}\big]}, \qquad \mathrm{CVA} = -(1-R_C)\int_0^T \lambda_C(s)\,D(0,s)\,\mathbb{E}\Big[\tfrac{\lambda_C(s,\omega)}{\lambda_C(s)}\,V(s)^+\Big]\,ds$$

Here β is the elasticity of the producer's default intensity to the oil price. β = 0 is Green's independence case; β > 0 is right-way risk.



Exposure profile of the oil hedge (Monte Carlo, 60,000 paths). The dashed line weights each path by the producer's default intensity at β = 1.

Step 4 — Capital and margin.

- SA-CCR EAD = 1.4(max(V, 0) + m · 18% · d · MF), recomputed on every path.
- d is the remaining barrels times the current oil price, and MF = √min(M, 1).
- K_CCR and K_CVA follow as in the KVA section, and KVA uses (13.25).
- MVA uses (10.34) on the expected initial margin of the futures hedge.
- EAD today is USD 70.2m, against an average EPE of USD 13.0m.

Results, independent credit (β = 0)

USD, present value	Value
Margin in the fixed price, V(0)	+1,693,368
CVA	−699,386
DVA	+282,225
FCA	−262,270
MVA on the futures hedge	−221,068
KVA on CCR capital	−751,745
KVA on CVA capital	−172,033
Total XVA	**−1,824,277**
Net to the bank	**−130,909**

Results with right-way risk. Only CVA moves. Capital, and so KVA, is unchanged, because SA-CCR and BA-CVA give no credit for right-way risk.

Elasticity β	CVA	Net to the bank
0 (independence)	−699,386	−130,909
1	−412,911	+155,566
2	−235,446	+333,031

What it shows

- **Right-way risk is the economics of hedging business.** A hedger's derivative offsets its underlying business, so the bank is owed most when the client is strongest. At β = 1, CVA falls 41% and the trade becomes profitable. Pricing at independence systematically overcharges hedgers.
- **It is not unconditional.**
    - Supply shocks are right-way: oil up, producers richer.
    - Demand shocks can be wrong-way. In 2020, oil and airline credit collapsed together, exactly when airline consumer hedges were in the bank's favour.
    - Producers who over-hedge relative to production carry specific wrong-way risk.
    - β should therefore be stressed by client type and calibrated as Green suggests: a stress parameter with governance, not a free benefit.
- **Capital is the binding cost.** KVA (USD 0.92m) exceeds CVA, and regulation ignores right-way risk. Without IMM approval, as under APRA, SA-CCR exposure (EAD 5.4× average EPE) prices the trade. An SA-CVA approval with eligible index hedges is the main lever on the CVA-capital part.
- **The hedge creates funding costs.** The client posts nothing, but the futures hedge needs initial margin (MVA USD 0.22m) and variation margin. Variation margin is a funding cost when oil rises and a benefit when it falls, and is captured in FCA and DVA through Green's funding constraint.
- **The mirror trade is a consumer.** For an airline buying oil forward, the exposure flips: the bank is owed when oil falls. CVA at independence is USD 0.75m, and it is right-way only for supply-driven price moves.

Hybrid models for the exposure engine (Ch. 16)

Philosophy.

- **Simple, Markovian, cacheable models, calibrated to market.** Green argues that richer models cost more in covariance management and calibration failures than they gain. Inputs such as proxy spreads are too uncertain for sub-basis-point exposure accuracy.
- **Factor counts.** Hull–White one-factor rates plus lognormal FX for GBP, EUR and USD with UK inflation is 7 factors. Two-factor rates with stochastic vol, stochastic basis and LSV FX is 27.

- **Market-implied calibration.** Required because XVA is part of fair value and hedges must reprice. Use historical vols only where no options trade, and historical correlations in practice.
- **Historical drift is dangerous.** Extrapolated, the 2008 GBPUSD fall puts the 30-year central path at about $0.0001.

Interest rates: Hull–White as the workhorse.

$$dr = \big(\theta(t) - a\,r\big)\,dt + \sigma(t)\,dW, \qquad P(t,T) = A(t,T)\,e^{-r(t)B(t,T)}, \qquad B(t,T) = \frac{1 - e^{-a(T-t)}}{a} \qquad (16.62,\ 16.65,\ 16.68)$$

Calibrate by bootstrapping a piecewise-constant σ(t) to at-the-money co-terminal swaptions, with mean reversion fixed historically or by judgement. Green prefers the Linear Gaussian Model with a long-bond numeraire (16.103, 16.167), which can be jointly calibrated to caplets and co-terminal swaptions. He rejects the LIBOR market model for XVA: it is non-Markovian, has no analytic bond price, and is hard to make multi-curve or hybrid.

Rate model	Volatility lines	Whole surface	Smile
Hull–White, LGM, G2++	Yes	No	No
Cheyette (quasi-Gaussian), quadratic Gaussian	Yes	Yes	Yes
LIBOR market model and variants	Yes	Yes	Partly, with SV extensions

Multi-currency. Each currency's short rate follows Hull–White; FX is lognormal with drift r_d − r_f. Under the domestic measure, the foreign rate picks up a quanto drift:

$$dr_f = \big(\theta_f - \alpha_f r_f - \rho_{X,f}\,\sigma_X\,\sigma_f\big)\,dt + \sigma_f\,dW_f^d \qquad (16.189)$$

$$\Sigma^2 = \sigma_X^2 + \sigma_f^2 B_f^2 + \sigma_d^2 B_d^2 + 2\rho_{X,f}\sigma_X\sigma_f B_f - 2\rho_{X,d}\sigma_X\sigma_d B_d - 2\rho_{f,d}\sigma_f B_f\,\sigma_d B_d \qquad (16.197)$$

σ_X is bootstrapped from at-the-money FX options; Green warns it can go negative at long maturities. Crosses obey the triangle rule (16.209):

$$\sigma_{gf}^2 = \sigma_{df}^2 + \sigma_{dg}^2 - 2\rho_{df,dg}\,\sigma_{df}\,\sigma_{dg}$$

An uncalibrated cross misprices, for example, a EURGBP cross-currency swap simulated under USD.

Other asset classes.

- **Inflation:** Jarrow–Yildirim with Hull–White dynamics (16.233–16.235).
- **Equity:** lognormal with a per-currency systemic factor (16.304).
- **Commodities:** simulate the forward curve directly (16.320).
- **Credit:** Gaussian or CIR intensity, or JCIR++ with jumps (16.352). Correlated diffusions give only weak wrong-way risk; a systemic jump is needed.

Monte Carlo, sensitivities and AAD (Ch. 17–21)

Every XVA is a strip of European-style options on the netting set (19.1–19.2). The engine therefore needs three things: an exposure distribution on a time grid, a valuation at each node, and a credit integration.

$$\mathrm{XVA}(t) = \int_t^T a(s)\, e^{-\int_t^s b}\, \mathbb{E}\big[(V(s) - X(s))^{\pm}\big]\, ds \;\approx\; \sum_i w(t_i)\, \mathbb{E}\big[(V(t_i) - X(t_i))^{\pm}\big]$$

Errors and random numbers.

- There are two discretisation errors: path stepping, and the time partition of the XVA integral. Only monthly or quarterly steps resolve a swap's saw-tooth exposure.
- Use exact SDE solutions where possible.
- Random numbers must be repeatable and use a tested generator; Mersenne Twister, or Sobol with a Brownian bridge.
- Never seed parallel streams differently from the same generator; use skip-ahead.
- Regularise correlation matrices to the nearest valid matrix before Cholesky.

Variance reduction. Per-time-step orthogonalisation (Curran) roughly halves the standard error, worth about 4× the paths. Antithetics and control variates add little on diverse books.

Valuation inside the simulation. The budget is about 100ms per trade valuation, and much less for most of the book. Green lays out the trade-off:

- Front-office-consistent pricers match spot exposure and smile.
- Model-consistent valuation, via Longstaff–Schwartz everywhere, is arbitrage-free and explains P&L cleanly.
- The valuation difference from front office should be monitored daily as a KPI. Scaling to the t = 0 price is allowed only under that monitoring.

Sensitivities. The second-order chain rule through the calibration Jacobian J and Hessian H (21.6–21.9):

$$\frac{\partial^2 \mathrm{XVA}}{\partial \bar{a}_i\, \partial \bar{a}_j} = \sum_k \frac{\partial \mathrm{XVA}}{\partial a_k}\, H^k_{ij} + \sum_{k,l} \frac{\partial^2 \mathrm{XVA}}{\partial a_k\, \partial a_l}\, J_{ki}\, J_{lj}$$

- **Finite differences.** Use common random numbers and test bump size against path count. Freeze the exercise decisions from the base run: a bumped path that stops exercising ("differential exercise") creates large noise.
- **Adjoint method.** Backward propagation through the path Jacobians gives all first-order sensitivities at about 3–4× the cost of one valuation, whatever the number of inputs. The cost is memory, especially on GPUs:

$$V(t_n) = D(t_n)^{\top}\, V(t_{n+1}), \qquad \frac{\partial g}{\partial X(t_0)} = V(t_0)^{\top} J(t_0) \qquad (21.31\text{–}21.35)$$

- **Green's preference.** A hybrid: analytic derivatives of the Longstaff–Schwartz regression functions, chained through the model's analytic Jacobians (21.42–21.44), with AD only where analytics run out.

P&L explain. The risk-based explain is a Taylor expansion with an unexplained residual ε (21.10). Carry artefacts have to be isolated: tenor-grid roll, random numbers shifting with the date count, and the CDS IMM roll.

Organisation. A single central XVA desk is "the optimal arrangement", with a common trade and market-data bus behind it. Green treats software governance as part of the model: test-driven development with numerical tolerances, peer review, source control, CI, and independent validation, possibly including a parallel implementation.

For a model owner: what to validate, re-derive and update

Green's framework is internally consistent, but every result depends on choices a validator should see documented. The most important is the funding strategy, because it decides which XVA terms exist.

Assumption	Where	Why it matters
Funding strategy (two bonds, no shortfall, one bond)	9.50–9.70	Decides whether DVA, FCA or FVA appear, and whether λB enters at all
Close-out convention (risk-free, risky, funding)	9.77–9.96	Risky close-out makes U non-linear; funding close-out separates CVA and FVA
No bond–CDS basis; λB from bonds vs CDS	9.16, 11.4	Sets the funding spread and whether DVA can be read as a funding benefit
Survival discounting at r + λB + λC on every term	All XVA integrals	Often dropped in production; changes KVA and MVA materially for weak names
Deterministic funding spread, hazard rates, recovery	Ch. 9–11	No funding or credit wrong-way risk
P-measure VaR shocks inside a Q-expectation	10.3.3	Measure mixing in MVA, and in KVA for internal-model capital
Capital as rented equity, φ = 0, constant γ_K	13.1–13.2	φ = 1 cuts KVA about 25%; γ_K is a board input, not a market observable
Capital allocation scheme	15.3	Counterparty KVA is not unique; only the total is
Hull–White/LGM with no smile; historical correlations	Ch. 16	Out-of-the-money exposure and vega hedges; cross-FX calibration
Regression basis and augmentation	10.35–10.40	Tail accuracy of IM and capital profiles; extrapolation risk

Apparent misprints to re-derive before implementing:

- **Ch. 3:** (3.25) sign of the repo term; (3.37) λ_C(u) should be λ_C(s); (3.47) bracket sign.
- **Ch. 8–9:** (8.30) missing minus in the exponent; (9.32) missing α_i; (9.36) sign; (9.61) sign; (9.82) exponent.
- **Ch. 13:** (13.15) undefined ψK term; (13.37), where r_B·φ is subtracted without the K factor and with r_B in place of r.
- **Ch. 16–17:** (16.201), (16.207); (17.7), (17.12), (17.50).

The FVA, MVA and KVA results in this doc, (9.55–9.59), (10.34) and (13.25), were checked against the page images and are as printed.

What is dated (book written 2014–15).

- **Rates:** LIBOR, EONIA and LIBOR–OIS basis throughout. Discounting is now on RFRs, and tenor basis is largely legacy.
- **Capital:** CEM, the old standardised CVA charge and IMM-based CVA capital are superseded by SA-CCR, BA-CVA and SA-CVA. FRTB appears in consultative form.

- **Margin:** bilateral IM appears before ISDA SIMM existed. Dynamic IM today is usually SIMM projected from sensitivities, not historical VaR.
- **Hardware:** Xeon Phi and K40 GPU timings.
- **Still current:** the replication framework, the integrals themselves, and the governance advice.

Links to the other two books.

- **Gregory.** Gregory's KVA and MVA formulas are this book's (13.25) and (10.34) in simplified form. Gregory's symmetric FVA is the one-bond strategy. Gregory states the FVA/DVA overlap; Green derives it.
- **Savine and Andreasen.** Their CVA as "cash flows discounted by a path-dependent credit process" is the same object as Green's credit-adjusted discounting. LSAC and their regression proxies are the same family of technique. Green's closing vision of one universal valuation model across pricing and XVA (Ch. 23) is what their scripting design implements.

For Macquarie.

- With no IMM approval under APRA, capital profiles are standardised.
- Green's §13.4 method fits that directly: compute SA-CCR EAD and BA- or SA-CVA capital as a payout in each Monte Carlo state, then integrate with (13.25).
- That puts the SA-CCR engine inside the KVA model, so one governed implementation should serve both.

Source: Andrew Green, *XVA: Credit, Funding and Capital Valuation Adjustments*, Wiley Finance, 2015, Chapters 1–23 (copy in the Kerr Shearer/Macquarie folder).